Xiangqi

Project Specification and Demo Guide

CS109 Introduction to Computer Programming Fall 2026

1. Introduction

In this project, you are required to implement the classic board game **Xiangqi** using **Java** . Players move their pieces to threaten the opposing General while protecting their own. Your project must include a graphical interface, complete gameplay rules, user accounts, and game saving/loading.

2. Board and Interface

The board has 9 columns and 10 rows of intersections. Pieces stand on the intersections. A river separates the two sides, and each side has a palace covering 3 columns and 3 rows of intersections.

A complete game starts with 16 pieces per side: 1 General (King), 2 Advisors, 2 Elephants, 2 Horses, 2 Rooks, 2 Cannons, and 5 Soldiers. Red starts at the bottom and Black at the top.

3. Game Rules

1. **Turns:** Red moves first. The players then alternate, moving exactly one piece per turn. Passing is not allowed.
2. **Moves and captures:** Follow the movement rules for each piece type. A piece may capture an opponent by moving to its intersection. A move cannot end on a friendly piece or leave the board.
3. **Check:** A General is in check when an opposing piece threatens to capture it. A legal move must leave the moving player's General safe. The two Generals must never face each other on the same column with no intervening piece.
4. **Game endings:**
   Win/loss: The player to move loses if none of their pieces has a legal move. In this situation, it is checkmate if that player's General is in check, and stalemate otherwise. A player who resigns also loses.
   Draw: the game is drawn when both players agree. Other draw rules are not part of the basic requirements.

Rule references:

- Xiangqi.com: Rules and Getting Started
- Xiangqi.com: Pieces and Moves
- World Xiangqi Federation: Introduction to Xiangqi

4. Project Requirements

Basic Tasks 1–5 total 80 points. **Advanced Features** completed in Week 15 are worth up to 30 points, while those completed in Week 16 are worth up to 20 points. **The final score is capped at 100 points.**

Task	Points
1. Game initialization	10
2. Multi-user login	15
3. Save and load games	15
4. Gameplay logic	30
5. Graphical user interface	10
6. Advanced features	30 (Week 15) / 20 (Week 16)

Task 1: Game Initialization (10 points)

1. Display the board correctly, including the river, palaces, and initial piece positions.
2. Reset the board to its initial state when starting a new game.
3. Allow players to start a new game at any time without restarting the program.

Task 2: Multi-user Login (15 points)

1. Provide registration and login interfaces.
2. Allow guests to play without logging in. Guests cannot save or load game progress.
3. Previously registered users must still be able to log in after the program is closed and reopened.

Task 3: Save and Load Games (15 points)

1. Allow registered users to save their current game progress, including the board state, player turn, and move history.
2. Provide at least one save slot per user. Allow a user to load a previous save from the menu.
3. Detect invalid formats or game data in corrupted or modified save files, and handle them appropriately without crashing.

Task 4: Gameplay Logic (30 points)

1. Implement legal movement rules for all seven piece types. Prevent illegal moves from being executed.
2. Implement capturing, check, checkmate, stalemate, and draws.
3. Provide an option to resign. Allow the game to end in a draw by mutual agreement.
4. Display appropriate messages for check, win/loss, and draw.
5. Display the previous move, for example by highlighting the moved piece or showing a textual record.

Task 5: Graphical User Interface (10 points)

1. Implement a graphical board and interactive piece movement using Swing, JavaFX, or another Java GUI framework.
2. Show whose turn it is and display check, win/loss, and draw states clearly. Normal gameplay should not require command-line interaction.
3. A command-line-only game will not receive full marks for this task.

Task 6: Advanced Features (Max 30 points in Week 15; 20 in Week 16)

Advanced or creative features may include, but are not limited to:

- An AI opponent with adjustable difficulty.

- Move suggestions, or legal-move highlighting.
- Timers or time-limited game modes.
- Move-by-move replay, undo, or move-record export and import.
- Importing custom positions or endgame setups.
- Online or local-network multiplayer.
- Automatic saving.
- Improved graphics, animations, board or piece themes, sound effects, or background music.
- An independently implemented GUI (not the one provided in the Demo).

5. Demo Guide

You may build on the provided Demo or implement your project from scratch. **The Demo is only a starting example; your submission must meet the project requirements above.**

Requirements and Launch

The Demo uses JDK 25, Maven, and Swing. Swing is included in the JDK; no third-party runtime library is required.

1. Extract the Demo archive to obtain the `XiangQi` folder.
2. In IntelliJ IDEA, choose File → Open and open the `XiangQi` folder containing `pom.xml`. Load it as a Maven project.
3. Open the following file and run its `main` method:
   `src/main/java/edu/sustech/xiangqi/Main.java`

中国象棋

项目说明与 Demo 使用指南

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1. 项目介绍

在本项目中，你需要使用 **Java** 实现经典棋盘游戏**中国象棋**。玩家通过移动棋子威胁对方的将帅，同时保护己方将帅。项目须提供图形界面、完整的对局规则、用户账号以及游戏存读档等功能。

2. 棋盘与界面

棋盘由 9 列、10 行交叉点组成。棋子放在交叉点上，河界分隔双方阵营，每方九宫覆盖 3 列、3 行交叉点。

完整对局中，每方初始有 16 枚棋子：1 个将／帅、2 个士／仕、2 个象／相、2 个马、2 个车、2 个炮和 5 个卒／兵。红方在下，黑方在上。

3. 对局规则

1. **回合：**红方先行，随后双方交替，每回合移动一枚棋子，不允许跳过回合。
2. **移动与吃子：**遵循各类棋子的移动规则。棋子可移动到敌方棋子所在的交叉点吃子；不能落在己方棋子上，也不能移出棋盘。
3. **将军：**敌方棋子威胁吃掉将帅时，称为将军。合法移动必须使己方将帅保持安全。双方将帅不得在同一列无棋子阻隔地直接照面。
4. **对局结束：**
   胜负：当前行棋方的所有棋子均无合法走法时，该方判负：若其将帅正在被将军，则为将死；否则为困毙。主动认输的一方也判负。
   和棋：双方同意后，对局判和（其他和棋规则不作为基础要求）。

规则参考链接：

- 相弈象棋：象棋规则
- 相弈象棋：棋子与走法
- 世界象棋联合会: Introduction to Xiangqi (In English)

4. 项目要求

基础任务 1–5 共 80 分。**高级功能**于第 15 周完成最高 30 分，第 16 周最高 20 分。**最终总分封顶 100 分**。

任务	分值
1. 游戏初始化	10
2. 多用户登录	15
3. 游戏存读档	15
4. 对局逻辑	30
5. 图形用户界面	10
6. 高级功能	30（第 15 周）/ 20（第 16 周）

任务 1：游戏初始化（10 分）

1. 正确显示棋盘，包括河界、九宫和棋子的初始位置。
2. 开始新游戏时，将棋盘恢复到初始状态。
3. 允许玩家随时开始新游戏，无需重启程序。

任务 2：多用户登录（15 分）

1. 提供注册和登录界面。
2. 游客无需登录即可游玩，但不能保存或读取游戏进度。
3. 程序关闭并重新运行后，已注册的用户仍可登录。

任务 3：游戏存读档（15 分）

1. 允许注册用户保存当前游戏进度，包括棋盘状态、当前回合和走棋历史。
2. 每个用户至少提供一个存档位，并允许从菜单读取此前保存的对局。
3. 检查损坏或被修改的存档中的格式、局面数据等错误，应妥善处理，避免程序崩溃。

任务 4：对局逻辑（30 分）

1. 实现全部七类棋子的合法移动规则，并阻止非法移动被执行。
2. 实现吃子、将军、将死、困毙以及和棋。
3. 提供认输选项，经双方同意可以和棋。
4. 显示相应的将军、胜负和和棋信息。
5. 显示上一步，例如高亮移动的棋子或显示文字记录。

任务 5：图形用户界面（10 分）

1. 使用 Swing、JavaFX 或其他 Java 图形界面框架，实现图形棋盘和交互式棋子移动。
2. 清楚显示当前回合、将军、胜负和和棋状态，正常游戏过程不应依赖命令行交互。
3. 仅有命令行界面的游戏将不能获得本项满分。

任务 6：高级功能（第 15 周最高 30 分；第 16 周最高 20 分）

高级或创意功能可以包括（但不限于）：

- 可调难度的人机对战。
- 走棋建议或合法落点高亮。
- 计时器或限时对局模式。
- 逐步回放、悔棋或棋谱导出与导入。
- 导入自定义局面或残局。

- 在线或局域网对战。
- 自动存档。
- 改进图形、动画、棋盘或棋子主题、音效或背景音乐。
- 独立实现的图形界面（而非使用提供的 Demo）。

5. Demo 使用指南

可以在提供的 Demo 基础上开发，也可以从头实现。**Demo 只是起步样例，提交的项目须完成上述要求**。

环境要求与启动

Demo 使用 JDK 25、Maven 和 Swing。Swing 随 JDK 提供，无需第三方运行时库。

1. 解压 Demo 压缩包，得到 `XiangQi` 文件夹。
2. 在 IntelliJ IDEA 中选择 File → Open，打开包含 `pom.xml` 的 `XiangQi` 文件夹，并作为 Maven 工程加载。
3. 打开以下文件，运行其中的 `main` 方法：
   `src/main/java/edu/sustech/xiangqi/Main.java`